

Nicholas D. Walker

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EDUCATION

Rensselaer Polytechnic Institute (RPI)

Dual B.S. in Mechanical Engineering and Aerospace Engineering
Minor in Economics

August 2021 – May 2025

Dean's List

Aerospace Specialty Courses GPA: 3.51

SKILLS AND COURSEWORK

Relevant Coursework: Linear Algebra & Differential Equations, Intro to FEA, Aerostructures, Aeroelasticity/Vibrations

Technical Skills: NX, MATLAB, Python, Arduino, SolidWorks, Autocad, 3Ds, CATIA, STK, LTSpice, C++, Linux

WORK EXPERIENCE

Hypersonic Flow Modeling Research, RPI

Troy, NY

Undergraduate Researcher

Jan 2025 – Ongoing

- Investigation of hypersonic shock wave boundary layer interactions to analyze effects on vehicle stability, heat transfer, and skin friction in high-speed applications
- Conducted high-fidelity OpenFOAM simulations with GMSH-meshed complex geometries for efficient modeling

Design Engineer Intern, GE Vernova

Schenectady, NY

Steam Turbine Design

Jan 2024 – April 2024

- Simulated turbine buckets using FEA (ANSYS), with mesh refinement study to validate convergence and accuracy for vibrational response and structural assessment
- Conducted root-cause analyses of control valve failures by integrating NX CAD models and GD&T from PLM, identifying critical design issues
- Prepared and delivered technical reports and presentations, using various MS Office products

Structural Damage Diagnostic Research, RPI

Troy, NY

Undergraduate Researcher

Sep 2023 – Dec 2023

- Developed numerical models for UAV damage state parameterization using a six degrees-of-freedom experimental setup; processed sensor data with MATLAB
- Designed FEM/SEM-based structural health monitoring (SHM) models to evaluate vibrational responses and failure modes in structure
- Performed computational and experimental fatigue analysis on vibrational data, collaborating with doctoral candidates to validate dynamic load models

PROJECTS

Europa Orbiter, RPI

Sep 2024 – Dec 2024

- Conducted orbital simulations and trajectory optimizations with STK, visualizing dynamic responses in MATLAB
- Modeled system components using Simulink and designed structural and propulsion systems in NX, addressing power distribution and load challenges in a multidisciplinary team

Yonka Rocket Design, RPI

Jan 2023 – May 2023

- Designed a carbon fiber rocket using CAD, NX and evaluated structural integrity through NASTRAN simulations
- Implemented Python-based numerical simulations (NumPy, SciPy) to model thrust profiles, flight stability, and aerodynamic drag, focusing on dynamic load assessments

Fil-A-Bot Sustainability Device, RPI

Jan 2023 – May 2023

- Developed a device to transform plastic bottles into 3D printing filament; programmed Arduino microcontrollers for real-time temperature and motor speed regulation with parts designed in SolidWorks
- Automated data analysis in Python to optimize extrusion parameters, ensuring uniform filament quality

LEADERSHIP EXPERIENCE

Rensselaer Rocket Society: RPI sponsored club purposed to design and build rockets using CREO, and CATIA

Lambda Chi Alpha Fraternity: Financial Advisor, reviewed the allocation of a \$10,000 budget

Bwenzi: Partnered with cross-cultural student leaders to raise funds for education aid in Malawi, aligned with UN Goals